

Primordial Black Holes as a Dark Matter Candidate

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(Dated: June 25, 2026)

Black holes are a good potential candidate for dark matter as they do not interact with light and exert a strong gravitational influence. While supermassive and stellar black holes fall short of providing a robust explanation of dark matter, there is another class of black holes formed from density perturbations in the early universe that could be the right contender. In this exploration, I will investigate the formation of these primordial black holes (PBHs) and attempt to explain how they can solve the dark matter problem. I will also detail our current knowledge of their abundance and their possible contribution to the dark matter density, as well as future experiments to keep updating this knowledge.

I. A BRIEF HISTORY OF DARK MATTER

Dark matter is one of the most confounding constituents of our universe, and although we understand a lot about its behavior, we do not really know much about its composition. What we do know is that it is **dark**, i.e. it does not interact electromagnetically (with light) and that it is **matter**, i.e. it exerts an enormous amount of gravitational influence.

But why was the existence of dark matter proposed in the first place? In 1933, Swiss astronomer Fritz Zwicky postulated the existence of “a significant amount of unseen or missing mass” (Vavryčuk, 2024) by attempting to compute the mass of the Coma galaxy cluster by measuring the velocity dispersion σ_v of its constituent galaxies (Koechler, 2024).

The virial theorem relates the average kinetic energy $\langle T \rangle$ and the average potential

energy $\langle U \rangle$ by $2\langle T \rangle + \langle U \rangle = 0$. Assuming that the galaxies are distributed uniformly in the cluster, modeled by a sphere of radius R , $\langle T \rangle = \frac{M\langle v^2 \rangle}{2}$, where M is its total mass and v is the velocity of the individual galaxies, and $\langle U \rangle = -\frac{3GM^2}{5R}$. Using these relations, we can calculate the predicted velocity dispersion $\sigma_v = \sqrt{\langle v^2 \rangle} = \sqrt{\frac{3GM}{5R}}$, which is approximately 80 km/s, using preexisting information of the mass and radius of the Coma cluster (Koechler, 2024). However, Zwicky measured this velocity dispersion to be around 1000 km/s (Koechler, 2024), much higher than expected, implying that the Coma cluster had a mass much larger than the mass of its visible constituents.

In 1970, Vera Rubin and Kent Ford measured the circular velocity v_c of stars in the Andromeda galaxy. By Newton’s second law, we know that this circular velocity should decrease with an increase in the square root of the distance from the center of the galaxy, $v_c \propto 1/\sqrt{r}$. However, Rubin and Ford ob-

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served that the circular velocities of stars in the outskirts of the Andromeda galaxy are approximately constant, implying the existence of a “halo of matter” that is invisible and interacts only gravitationally and “extends beyond the galactic disk” (Koechler, 2024).

Since then, we have measured the abundance of dark matter with “exquisite precision” (Bird et al., 2016), but its “nature and identity” remains unknownst to us (Felderke and Sibiryakov, 2024). We have many ideas for what it might be, involving many modifications to current accepted theory, but there is one candidate which “fulfills all the necessary requirements” (Green, 2024), and does not need any new physics — black holes.

II. BLACK HOLES AS DARK MATTER

A black hole is a region of spacetime where the gravitational field is “so strong that not even light can escape” (Carr and Kühnel, 2022). At first glance, these seem to be ideal candidates to explain dark matter — they have mass to interact gravitationally and cannot be detected visually, only through their gravitational effects on matter or the radiation emitted by their accretion disks. In fact, according to Bird et al., “[c]ompact objects, particularly black holes, are one of the oldest models of dark matter” (2023).

Fig. 1 shows a picture of Sagittarius A*, the supermassive black hole in the center of our galaxy, with a mass of $4 \times 10^6 M_\odot$. This image was captured by the Event Horizon

Telescope (EHT) in 2022, and is, in reality, an image of the black hole’s “silhouette” against its “glowing surroundings” (Lutz, 2024), not an image of the black hole itself. Among other challenges the EHT team had to face, the black hole is virtually invisible, showing its strength as a dark matter candidate.

Supermassive black holes (SMBHs) like these have masses ranging from $10^6 M_\odot$ to $10^{10} M_\odot$ (Carr and Kühnel, 2022). With these incredibly large masses, SMBHs tend to fall to the bottom of gravitational potential wells in galaxies and are “thought to reside in galactic nuclei” (Carr and Kühnel, 2022). Because of this, they cannot explain the dark matter “halo” inferred around galaxies by the likes of Zwicky, Rubin, and Ford, as well as many other experiments.

Stellar black holes formed from super-

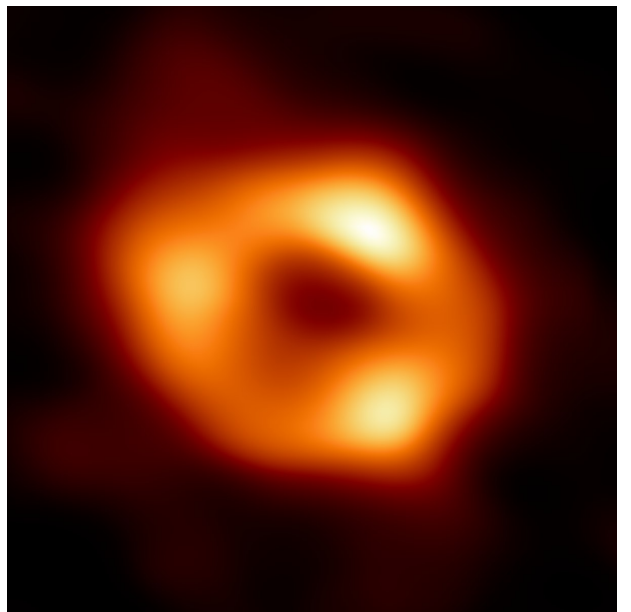


FIG. 1. Supermassive black hole Sagittarius A*, located in the Milky Way galactic nucleus. Event Horizon Telescope, 2022

novae, the “collapse of a massive star” (Villanueva-Domingo et al., 2021), at the end of its lifetime, have a minimum mass of around $3M_{\odot}$, set by the Tolman-Oppenheimer-Volkov limit (Villanueva-Domingo et al., 2021). Due to their lower mass, their concentration is more spread out and multitudes of them could exist in the “disk of our own [g]alaxy” (Carr and Kühnel, 2022). However, we don’t observe enough supernovae or heavy metal abundance (the products of supernovae) to infer a high enough stellar black hole concentration to explain the dark matter halo. Additionally, we can measure the density of dark matter from CMB anisotropies (Zaldarriaga, 2004), so we know that it has been present since long before the first stars were formed.

However, Zel’dovich and Novikov (1967) and Stephen Hawking (1971) showed independently that “gravitationally collapsed objects of very low mass”, or black holes, could have been formed in the early, radiation-dominated universe after inflation. These “primordial” black holes could still be around today and remain a good candidate to explain dark matter.

III. PRIMORDIAL BLACK HOLES

Black holes are formed when matter is compressed into such a dense and compact form that it collapses in on itself gravitationally. For matter of mass M to form a black hole, it needs to be squeezed into a region of space with a radius R_s , called the Schwarzschild radius, where $R_s = \frac{2GM}{c^2}$ (Carr and Kühnel, 2022). This matter could be

compressed to form a black hole in a variety of ways, such as the gravitational collapse of the core of a fuel-depleted star in the absence of any outward resistive force due to nuclear fusion, in the case of stellar black holes.

Primordial black holes (PBHs), on the other hand, are primarily formed from the “collapse of large density perturbations during radiation domination” (Green, 2024). In the early universe after inflation, overdensities entered the horizon as it grew. If these overdensities were large enough, there could potentially be enough mass in the volume of the horizon to collapse gravitationally into a black hole. Green states that these overdensities are the “same density perturbations that on large scales form galaxies and large-scale structure” (2024), but on smaller scales (smaller wavenumber k), with much higher amplitude to have sufficient mass to collapse gravitationally.

More specifically, for primordial black holes to form from an overdensity entering the horizon, the overdensity $\delta = \frac{\rho - \bar{\rho}}{\bar{\rho}}$, where ρ is the density of the region and $\bar{\rho}$ is the mean density of the universe, must be greater than some critical value δ_c , which is of the order of the equation of state parameter $w = 1/3$ (Carr, 1975).

Thus, the mass of the PBH formed is of the order of the mass in the horizon at the time the overdensity enters (Green, 2024), scaled by some proportionality factor γ which “depends upon the details of the gravitational collapse” (Villanueva-Domingo et al.,

2021), telling us that

$$M_{PBH} = \gamma M_H \sim \gamma 10^{15} \left(\frac{t}{10^{23} \text{s}} \right) \text{g} \sim \gamma \left(\frac{t}{10^{-6} \text{s}} \right) M_\odot \quad (1)$$

where M_H is the mass contained within the horizon at time t , and M_{PBH} is given in both grams g and solar masses $M_\odot$. This gives us an “enormous mass range” for PBHs, from having the Planck mass (10^{-5}g) if they were formed at Planck time (10^{-43}s), to a whopping $10^5 M_\odot$ if they were formed at one second after the Big Bang (Carr and Kühnel, 2020). This makes them solid candidates for black hole dark matter.

Assuming that the probability distribution $P(\delta)$ of overdensities is Gaussian, as shown in Fig. 2, we can estimate the initial abundance β for a particular PBH mass at the moment of formation (Villanueva-

Domingo et al., 2021), given by

$$\beta(M_{PBH}) \simeq \gamma \int_{\delta_c}^{\infty} P(\delta) d\delta \quad (2)$$

$$\simeq \gamma \sqrt{\frac{2}{\pi}} \frac{\sigma(M_{PBH})}{\delta_c} \exp\left[-\frac{\delta_c^2}{2\sigma^2(M_{PBH})}\right]$$

where $\sigma^2(M_H)$ is the mass variance.

Considering an “initial scale invariant power spectrum, at CMB scales”, the fluctuation amplitude $\sigma(M_{PBH})$ is around 10^{-5} , which gives an initial abundance of $\beta \sim 10^{-5} \exp(-10^{10})$ (Villanueva-Domingo et al., 2021), which is incredibly low. However, we must remember that this calculation assumes a Gaussian probability of overdensities. Villanueva-Domingo et al. also argue that non-gaussianities could have a “great impact on the initial fraction” (2021), which would depend heavily on the tail of the distribution. They also state that, “since matter and radiation densities scale differently with redshift” (2021), an initially small abundance β of PBHs could still be significant today.

The shape of the inflaton potential can result in different mass functions for the formed PBHs. A “monochromatic” mass function, where all PBHs formed are of the same mass, can be generated by an inflatory model called “chaotic new inflation”, resulting in sharp peaks in the primordial power spectrum (Villanueva-Domingo et al., 2021). On the other hand, “extended” mass functions, which “span over a large range of PBHs masses”, could be generated by other inflatory models like “hybrid inflation” (Villanueva-Domingo et al., 2021). For the sake of simplicity, I have only considered

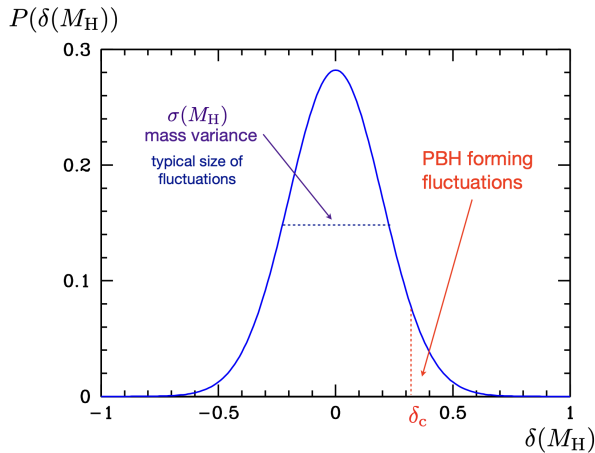


FIG. 2. Gaussian probability distribution $P(\delta(M_H))$ for primordial overdensities δ in terms of the mass M_H in the horizon. Primordial black holes form when overdensities exceed a critical value $\delta_c \simeq 1/3$. Green, 2024

a monochromatic mass function but will also discuss the effects of an extended mass function in the next section.

PBHs can also be formed by a “first-order phase transition” of a scalar field from a false vacuum to a true vacuum in the early universe (Kawana and Xie, 2021). According to Kawana and Xie, “this mechanism consists of a real scalar ϕ and a Dirac fermion χ ” (2021). The scalar field transitions to the new (true) vacuum from the old (false) vacuum through the formation of “bubbles”, and the fermions, which “cannot penetrate into the new (true) vacuum bubbles”, are “trapped in the old (false vacuum” (Kawana and Xie, 2021). As the phase transition takes place, the true vacuum bubbles grow and the false vacuum shrinks, compressing the fermions into “macroscopic compact objects” called Fermi-balls, which “collapse into PBHs when the internal Yukawa force becomes dominant” (Kawana and Xie, 2021). Kawana and Xie also detail PBH formation mechanisms through “the collapse of cosmic topological defects [and] scalar field fragmentation” (2021).

IV. NARROWING DOWN THE MASS RANGE

Now that we know that PBHs could take on a variety of masses, from very small to very large, we must experimentally undertake the task of ruling out the contribution of particular PBH masses to dark matter. By observing the universe, we can infer the concentrations of PBHs of specific masses,

and compare this to the known density and concentration of dark matter in the universe, allowing us to judge the validity of PBHs as a dark matter candidate.

Fig. 3 shows the current experimental constraints on $f_{PBH} = \rho_{PBH}/\rho_{DM}$, the fraction of dark matter that could be made up of PBHs, as a function of PBH mass, assuming a monochromatic PBH mass function.

The **orange** region shows constraints on f_{PBH} due to PBH **evaporation**. Black holes radiate their mass away due to a phenomenon called Hawking radiation (Villanueva-Domingo et al., 2021) and thus evaporate with a lifetime $\tau \propto M^3$, where M is the mass of the black hole (Carr and Kühnel, 2020). For black holes with mass $M \lesssim 5 \times 10^{14} \text{g}$, this lifetime τ is less than the age of the universe (Carr and Kühnel, 2020). So, any primordial black hole formed in the early universe, right after inflation, with a mass of around $10^{-18} M_\odot$ or less would have evaporated by now. Black holes of these masses could not, therefore, explain the density of dark matter, as they do not exist anymore to exert any gravitational influence. So, our mass spectrum begins from $M_{PBH} = 10^{15} \text{g}$.

PBHs with masses slightly above this critical mass must be “evaporating at a significant rate” (Green, 2024), and we should be able to detect them through the “strong γ ray and cosmic ray background” (Villanueva-Domingo et al., 2021) that they emit. We can infer their abundance by looking for this background. Villanueva-Domingo et al. state that the “absence of its detection strongly constrains the range

of masses $M_{PBH} \lesssim 10^{17}g$ ” (2021), as the orange section in Fig. 3 shows us.

The **blue** region shows constraints on f_{PBH} from observations of gravitational **microlensing**. MACHOs, massive compact halo objects (Carr and Kühnel, 2020), which include PBHs, can be detected by their lensing effects on distant light sources, as the curved spacetime around them amplifies the flux of the light source as it passes in front of it (Villanueva-Domingo et al., 2021). The frequency and intensity of these lensing events gives us information about the concentration and masses of the objects causing them. These microlensing effects are produced by “[p]lanetary and [s]olar mass PBHs (Green, 2024), and we can constrain this section of the mass range by observing these effects.

The MACHO Project searched for microlensing in the Large Magellanic Cloud

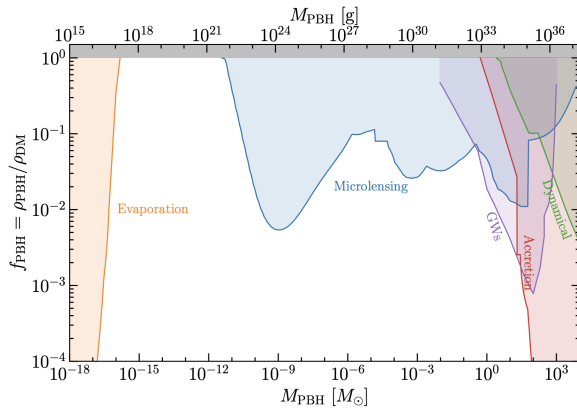


FIG. 3. Constraints on f_{PBH} , the fraction of dark matter that could be formed by PBHs, as a function of PBH mass. Orange, blue, purple, red and green regions show PBH masses ruled out from contributing to f_{PBH} by evaporation, microlensing, gravitational waves, accretion, and dynamical constraints respectively. Green, 2024

(LMC) for 5.7 years, detecting 13-17 such events (Alcock et al., 2000). These events were caused by MACHOs with mass $M \sim 0.5M_{\odot}$, and the MACHO project concluded that compact objects like PBHs could contribute at most 10% of the mass of the dark matter halo (Carr and Kühnel, 2022). Other microlensing projects, like EROS and OGLE, added to this to establish the blue constraint in Fig. 3, where $f_{PBH} \lesssim 0.01 - 0.1$ for this mass range.

The **purple** region shows constraints on f_{PBH} from the detection of **gravitational waves**. A gravitationally bound black hole binary will, over time, radiate energy in the form of gravitational waves, causing the black holes to “rapidly spiral inward” and “ultimately merge” (Bird et al., 2016). In 2015, LIGO detected gravitational waves from the merging of two black holes of mass $M \sim 30M_{\odot}$ (B.P. Abbott et al., 2016). Since these black holes are larger than the expected mass of a stellar black hole, set by the Tolman-Oppenheimer-Volkov limit, there is a possibility that these black holes might be PBHs (Bird et al., 2016). The black hole merger rate observed by LIGO and Virgo can help us set constraints on the concentration of multi-solar mass black holes, allowing us to determine their contribution to dark matter. According to Green, LIGO and Virgo would have to observe “several orders of magnitude more merger events” (2024) in order for all of dark matter to be made of these multi-solar mass PBHs. Villanueva-Domingo et al. state that the observed merger rates set “tight upper bounds of $f_{PBH} \lesssim 0.01$ ” on PBH masses

“between 1 and $300M_{\odot}$ ” (2021).

Additionally, if there was a sizable concentration of massive PBHs, we would expect to observe a “stochastic gravitational wave background” (Villanueva-Domingo et al., 2021), which also helps us constrain f_{PBH} (Carr and Kühnel, 2020). All these observations help us to exclude the gray region in Fig. 3.

The **red** region shows constraints on f_{PBH} due to the **accretion** effects of PBHs. Large black holes accrete matter, which gains energy and emits radiation as it falls into the black hole. By observing the “consequences of their accretion” (Green, 2024), we can constrain multi-solar mass PBHs. These accretion consequences can be observed by their contribution to “present day X-ray or radio emission” (Green, 2024) as well as their effects on the temperature anisotropies of the CMB (Villanueva-Domingo et al., 2021), allowing us to infer PBH concentrations. Since multi-solar mass PBHs “could have a high luminosity” (Carr and Kühnel, 2020) due to accretion, looking for these accretion consequences helps us constrain f_{PBH} , as shown in the red region in Fig. 3. Carr and Kühnel also note that analyzing PBH accretion is “complicated because the black hole luminosity will generally boost the matter temperature of the background Universe” (2020), adding further complications to the determination of accretion constraints.

Finally, the **green** region shows constraints on f_{PBH} from the observation (or lack thereof) of **dynamical** effects that massive PBHs must have on astrophysical systems, like stars “in wide binaries and

dwarf galaxies” (Green, 2024). A high enough concentration of massive PBHs could “dynamically heat star clusters”, causing the clusters to expand and eventually dissolve as their constituent stars gain energy due to the virial theorem (Villanueva-Domingo et al., 2021). Due to similar effects, ‘wide star binaries are “particularly vulnerable to disruption by PBHs” (Carr and Kühnel, 2020). Additionally, the “dynamical friction of various stellar populations” should cause halo objects like PBHs to be “dragged into the nucleus of the [g]alaxy” generating a higher mass in the galactic nucleus than that observed, allowing us to constrain the number density of PBHs (Carr and Kühnel, 2020). A high enough concentration of PBHs can “overheat the stars in the [g]alactic disc” (Carr and Kühnel, 2020), causing an increase in their vertical velocities and a thickening of the galactic disc, and the non-detection of this effect adds to our constraints in green.

It is also interesting to note that much smaller PBHs of mass $10^{19}\text{g} \lesssim M \lesssim 10^{22}\text{g}$ must also have dynamical effects on stars, although Carr and Kühnel state that the constraints due to these effects are controversial (2020) and are thus not shown in Fig. 3. If a small PBH passes through a white dwarf star, it causes “localized heating” in the white dwarf through “dynamical friction”, generating a type Ia supernova (Carr and Kühnel, 2020). The observed number and frequency of these supernovae excludes this PBH mass range from providing a significant contribution to the dark matter density.

As mentioned earlier, the constraints in Fig. 3 have been calculated assuming that

the PBH mass function is monochromatic. Realistic PBH mass functions, however, are not expected to be monochromatic even if all PBHs form at the same time, as “critical collapse means that they will have a range of masses” (Green, 2024). Taking this into account, the constraints shown in Fig. 3 will be “smeared out” as the constraint now applies “over a wider range of peak masses”, even though the tightest value of the constraint becomes weaker (Green, 2024).

Fig. 3 also shows an unconstrained region with mass $10^{17}\text{g} \lesssim M \lesssim 10^{22}\text{g}$, known as the “asteroid mass window” (Green, 2024), where PBHs could contribute to all of the dark matter density, i.e. $f_{PBH} = 1$. PBHs in this window are hard to detect because

of their low mass. Future experiments will attempt to tackle this range and update existing constraints.

According to Green, “future MeV gamma-ray telescopes will tighten the evaporation constraints”, while a “future microlensing survey of white dwarfs in the Large Magellanic Cloud” could tighten the microlensing constraints (2024). Both of these will eat into the fringes of the asteroid mass range, but new techniques will be required to probe the rest of this mass range. The microlensing of gamma-ray bursts can help us glean information about the number density of asteroid mass PBHs, with recent work showing “exciting potential” (Fedderke and Sibiryakov, 2024). In addition to this, “microlensing of X-ray pulsars” can also help tackle this range (Green, 2024). Fig. 4 shows the current and predicted constraints on f_{PBH} in the asteroid mass range.

V. CONCLUSION

We don’t have enough evidence to prefer PBHs over any other dark matter candidate, and no reason to disregard them completely. All we can do is keep chipping away, innovating with new experiments, and hope one day to hit gold. For PBHs, our goal right now is to tackle the asteroid mass range. If we can constrain f_{PBH} enough in this range as predicted by Fig. 4, using the methods detailed in the previous section, we can rule out PBHs from contributing to the entire dark matter density, i.e. $f_{PBH} \neq 1$ for any PBH mass.

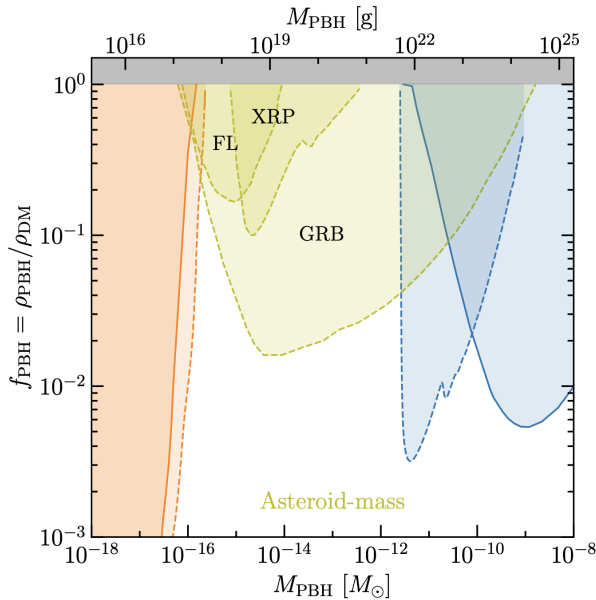


FIG. 4. Solid regions show current constraints, while dotted regions show projected future constraints on f_{PBH} in the asteroid mass range. Yellow dotted regions show predicted constraints from gamma-ray bursts (GRB), X-ray pulsars (XRP), and femtolensing (FL). Green, 2024

Even then, this does not mean that PBHs are not dark matter — they can still contribute to a part of the dark matter density. This would lend itself to a hybrid dark matter model, where the mysterious invisible mass exerting its gravitational influence on the universe is made up of many different constituents, like particles, PBHs, and other candidates.

Even if we can rule out PBHs as dark matter, studying them can help us glean important information about our universe. Detecting and quantifying their number density and mass range can tell us about the shape of the inflaton potential that formed them. Their abundance can help us constrain and reconstruct the primordial power spectrum (Wang et al., 2023),

and also explain the baryon asymmetry of the universe (Kawana and Xie, 2021). Additionally, PBHs can act as seeds for supermassive black holes in the first gigayear of the universe, as the presence of SMBHs at redshift $z \gtrsim 6$ “challenges current models of BH formation and evolution” (Ziparo et al., 2024).

Whatever we may learn from studying them, we know for a fact that PBHs play an important role in our universe, considering that they were formed at all. Ultimately, PBHs are a testament to how much knowledge there is for us to discover, and if we do confirm their existence experimentally, it might just be one of humanity’s greatest achievements.

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